

Hybrid Solar Charger

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Efficiency of a solar charging system depends on the weather conditions. Usually the solar panel gets four to five hours of bright sunlight in a day. If the weather is cloudy or rainy, it affects the charging process and the battery does not attain full charge. This simple hybrid solar charger can solve the problem as it can charge the battery using both



solar power as well as AC mains supply. When output from the solar panel is above 12 volts, the battery charges using the solar power. When the output drops below 12 volts, the battery charges through AC mains supply. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows circuit for the hybrid solar charger, which is built around a 12V, 10W solar panel (connected at SP1).

operational amplifier CA3130 (IC1), transistor BC547 (T1), 12V single-changeover relay (RL1), step-down transformer X1 and a few other components.

In bright sunlight, the 12V, 10W solar panel provides up to 17 volts DC with 0.6-ampere current. Diode D1



Fig. 1: Author's prototype

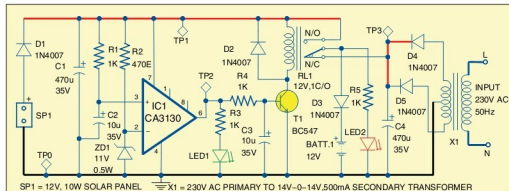


Fig. 2: Hybrid solar charger circuit

PARTS LIST

Semiconductors:

- | | |
|-------|--------------------------|
| IC1 | - CA 3130 op-amp |
| D1-D5 | - 1N4007 rectifier diode |
| LED1 | - 5mm green LED |
| LED2 | - 5mm red LED |
| ZD1 | - 11V, 0.5W zener diode |
| T1 | - BC547 npn transistor |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|-----------|--------------|
| R1, R3-R5 | - 1-kilo-ohm |
| R2 | - 470-ohm |

Carnicitors:

- C1, C4 - 470uF, 35V electrolytic
C2, C3 - 10uF, 35V electrolytic

Miscellaneous:

- | | |
|--------|---|
| BATT.1 | - 12V, 4.5Ah or 7Ah
rechargeable battery |
| SP1 | - 12V, 10W solar panel |
| RL1 | - 12V, 1C/O relay |
| X1 | - 230V AC primary to
14V-0-14V, 500mA
secondary transformer |

Test Points

Test point	Details
TP0	0V, GND
TP1	Solar panel voltage (12V to 17V depending on sunlight's intensity)
TP2	High (when solar voltage above 12 V)
TP3	15V to 16V when mains is connected

provides reverse polarity protection and capacitor C1 buffers voltage from the solar panel. IC1 is used as a simple voltage comparator. Zener diode ZD1

provides a reference voltage of 11 volts to the inverting input of IC1, while the IC's non-inverting input gets voltage from the solar panel through R1.

Working of the circuit is simple. When output from the solar panel is 12 volts or more, zener diode ZD1 conducts and provides 11 volts to the inverting terminal of IC1. Since its non-inverting

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